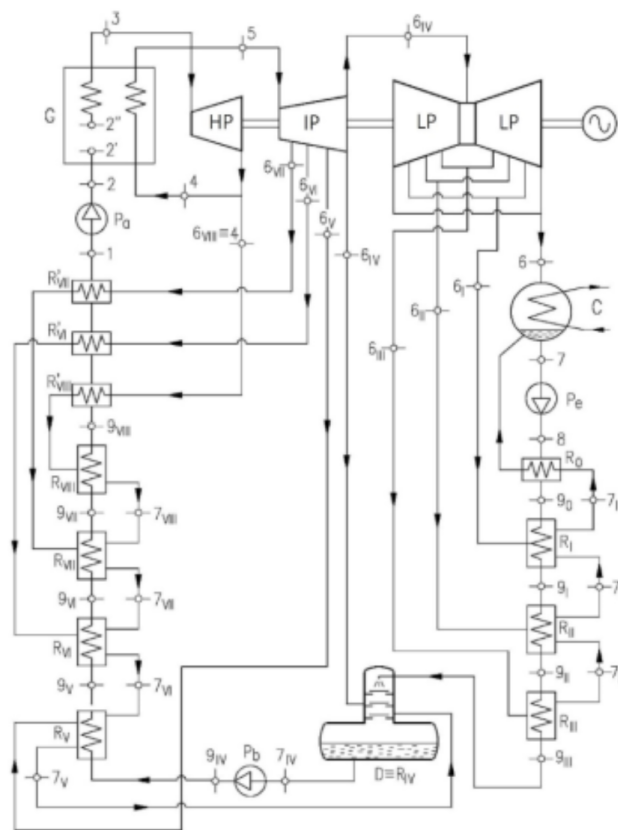


Homework 3 - Steam Turbines

LELME2150: Thermal Cycles



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Introduction

As part of the **Thermal cycles** course, we were asked to model an advanced steam turbine cycle. This report first presents the main assumptions and methodology used, then presents the results obtained in default operating conditions. Finally, a parametric analysis is performed to explore potential cycle improvements and efficiency gains.

1 General methodology and assumptions

1.1 Model assumptions

The **main** assumptions used to model the steam turbine cycle are the following :

- The combustion process is modelled as a simple adiabatic heat transfer.
- The pressure losses in the steam generator are taken into account.
- Isobaric condensation.
- Adiabatic expansions in the turbines (the isentropic model is used).
- The pumps are electrically powered rather than mechanically driven by the same shaft as the turbines.
- For the exergy analysis, the expansion valves between each feedwater heater are neglected.

1.2 Methodology

In order to develop the model, we first computed the thermodynamic quantities for all states of the cycle. Based on these quantities, we calculated the different flow rates in the bleedings through energy balances at each feedwater heater. The total flow rate of the cycle was determined using the power rating. Finally, the energy and exergy losses were calculated using energy and exergy balances on each component of the cycle, and the energetic and exergetic efficiencies were determined.

2 Default conditions results

In this section, we will review the results obtained with the model under default conditions. Those conditions are synthesized hereafter :

- $P_e = 288MW$
- $\eta_{is_IP} = 0.90$
- $T_{cd_out} = 305.15K$
- $\eta_{pump} = 0.85$
- $\eta_{is_HP} = 0.92$
- $T_{cd_subcool} = 1K$
- $\eta_{mec} = 0.98$
- $T_{drum} = 421.85K$
- $p_3 = 31MPa$
- $\eta_{is_LP} = 0.90$
- $T_{max} = 838.15K$
- $p_4 = 7MPa$

2.1 Table of the states obtained

States	$t [^{\circ}C]$	$p [kPa]$	$x [-]$	$h [kJ/kg]$	$s [kJ/kgK]$	$e [kJ/kg]$
1	285.8	10000	-	1265.6	3.112	370.6
2	293.5	35000	-	1295.1	3.105	401.9
3	565.0	31000	-	3320.7	6.077	1571.1
4	330.8	7000	-	2955.5	6.130	1190.6
5	565.0	6200	-	3574.7	7.056	1543.0
6 _{VII}	485.0	3751	-	3414.7	7.080	1376.2
6 _{VI}	404.0	2148	-	3254.7	7.106	1208.6
6 _V	321.8	1147	-	3094.7	7.136	1039.9
6 _{IV}	238.4	558	-	2934.6	7.172	869.7

States	t [°C]	p [kPa]	x [-]	h [kJ/kg]	s [kJ/kgK]	e [kJ/kg]
6 _{III}	154.1	239	-	2774.6	7.214	697.6
6 _{II}	95.6	86	0.98	2614.6	7.262	523.7
6 _I	66.8	27	0.93	2454.6	7.314	348.6
6	33.0	5	0.89	2294.5	7.521	128.9
7	32.0	5	0.00	134.1	0.464	1.9
9 _{VII}	246.6	10000	-	1069.4	2.748	279.2
9 _{VI}	216.0	10000	-	927.8	2.467	218.5
9 _V	185.9	10000	-	793.6	2.184	165.9
9 _{IV}	149.9	10000	-	637.8	1.831	111.9
7 _{IV}	148.7	460	0.00	626.6	1.829	101.3
9 _{III}	125.9	460	-	529.1	1.591	72.2
9 _{II}	95.6	460	-	400.7	1.257	40.3
9 _I	66.8	460	-	279.9	0.915	17.8
8	32.0	460	-	134.6	0.464	2.4

2.2 T-s and h-s diagrams

The T-s and h-s diagrams in default operating conditions are shown in figures 1 and 2 respectively.

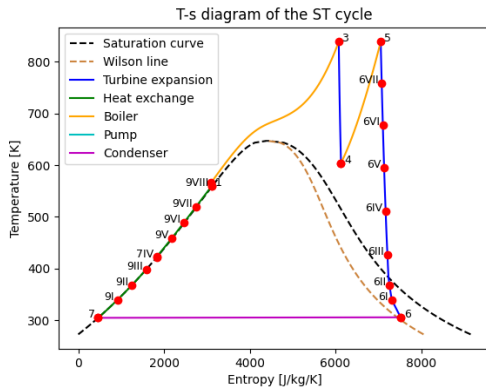


Figure 1: T-s diagram

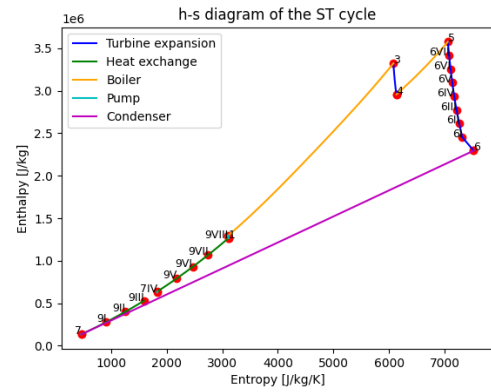


Figure 2: h-s diagram

2.3 Energy and exergy pie charts

The energy and exergy pie charts in default operating conditions are shown in figures 3 and 4 respectively.

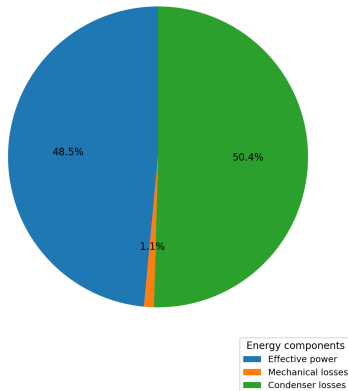


Figure 3: Energy pie chart

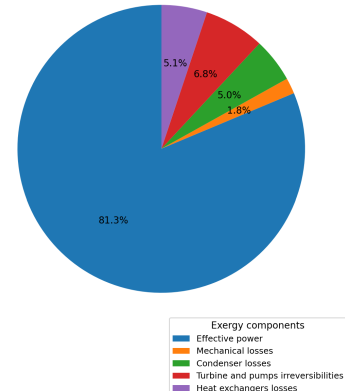


Figure 4: Exergy pie chart

3 Parametric analyses

In this section, we will analyze how several model parameters affect the system's performance. First, we will examine the effect of the number of bleedings. Next, we will analyse the effect of the number of reheatings. Finally, the effect of the pressure at the high-pressure turbine inlet will be discussed.

3.1 Influence of the number of bleedings

To analyse the influence of the number of bleedings, we varied the number of bleedings in the low-pressure turbine from 0 to 5. With the number of bleedings in the intermediate-pressure and high-pressure turbines kept constant, this adjustment allowed the total number of bleedings to range from 5 to 10. Our results are presented in figure 5.

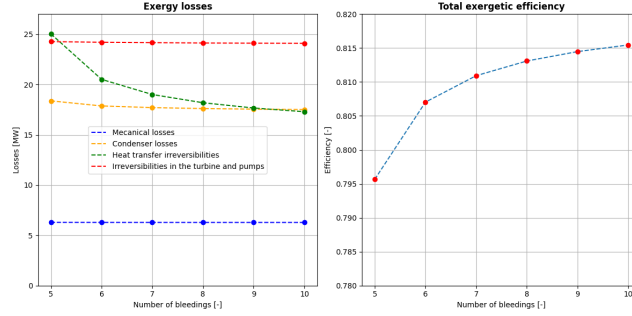


Figure 5: Influence of the number of bleedings on the system's performance.

Analysis : As expected, the total exergetic efficiency increases with the number of bleedings but this increase is not linear and tends to saturate. At one point, the costs related to the installation of an additional bleeding will exceed the benefits in terms of efficiency. This benefit can be explained by three main factors :

1. When a bleeding is extracted from the turbine, the mass flow rate of steam entering the condenser is reduced. This reduces the exergy losses at the condenser (p_{condex}).
2. The losses due to irreversibilities in the heat exchangers ($p_{transex}$) are reduced because the extracted steam preheats the feedwater in stages, lowering the temperature difference and minimizing exergy destruction.
3. The mechanical losses (p_{mec}) and the losses due to irreversibilities in the turbine and pumps (p_{rotex}) remain more or less constant.

Consequently, as the number of bleedings increases, the overall exergy destruction in the cycle is reduced.

Remark : For this analysis, the total exergetic efficiency (η_{totex}) is defined as follows ;

$$\eta_{totex} = \frac{P_e}{P_e + p_{transex} + p_{rotex} + p_{condex} + p_{mec}}$$

we obtain relatively high values ($\approx 80\%$) because the chimney losses, combustion irreversibilities and heat transfer irreversibilities in the steam generator are not considered.

3.2 Influence of the number of reheatings

In this part, we will observe the influence of the number of reheating systems on the cycle. We will use the default conditions to compare the T-s diagrams of the cycle with zero reheating, one (see Fig. 1) and two reheating systems. The T-s diagrams of the zero reheating and two reheatings are shown in figures 6 and 7 respectively. First, we know, for the same conditions, that the zero reheating system won't be possible to implement. Indeed, the vapor quality at the low-pressure turbine outlet is inferior

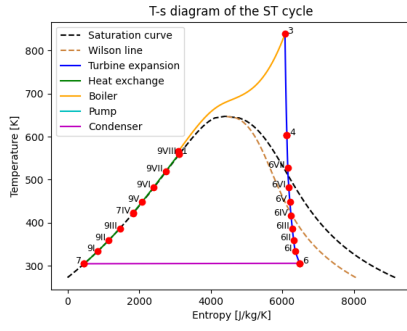


Figure 6: T-s diagram for zero reheating

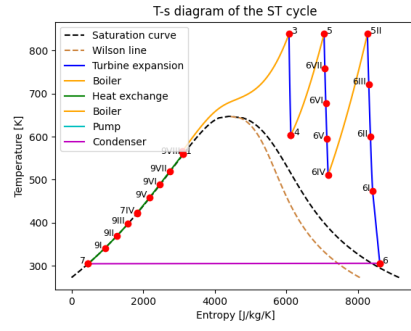


Figure 7: T-s diagram for two reheatings

to the limit imposed by the Wilson line. Then, we will discuss over the two reheatings system. This cycle is as efficient as the default one (0.1 % difference). However, we can see that the vapor at the low-pressure turbine outlet is still overheated. Therefore, the mass flow rate of cooling fluid passing through the condenser is higher due to the higher enthalpy of the point 6.

3.3 Influence of the pressure at the high-pressure turbine inlet

As final analysis, we will analyse the effect of the pressure at the high-pressure turbine inlet on the cycle.

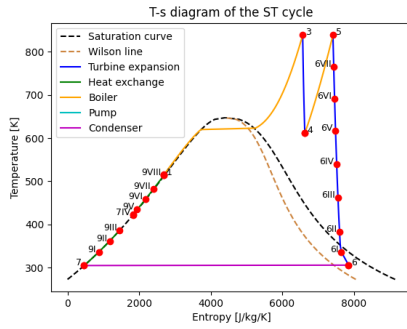


Figure 8: T-s diagram for $p_3 = 15000$ kPa

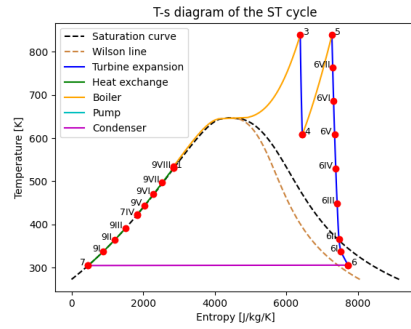


Figure 9: T-s diagram for $p_3 = 20000$ kPa

We can see in figure 8 that the pressure p_3 is under critical since we have a two-phase mixture in the boiler. In figure 9, the pressure is near the critical point. In the energy analysis, we can show that by increasing the pressure, we increase the energy efficiency of the cycle (and the exergy efficiency). As p_3 increases from 15,000 kPa to 31,000 kPa, the total energy efficiency rises from 0.461 to 0.485. By using the supercritical cycle, the boiler doesn't need to provide latent heat to water. The drawback is that the infrastructure cost is higher (pump, materials, etc.).

Conclusion

As conclusion, we can say that the numbers of bleedings and reheatings have a significant impact on the cycle's efficiency. These numbers can vary depending on the application and the constraints. We also see that the Rankine cycle can be improved by using supercritical pressures. Compared to a gas power plant, the efficiency is higher but the constraints are more important. Indeed, we need a source of cooling fluid and so, another cycle designed to manage the heat discharged by the condenser. Even though the Rankine cycle has been used for a long time with non-renewable sources, it can still be used in more modern applications. We can think of combined cycle power plants or ROC's (Rankine Organic Cycles) which create electricity from heat waste or renewable sources.